

Enterprise Architecture

UTHM

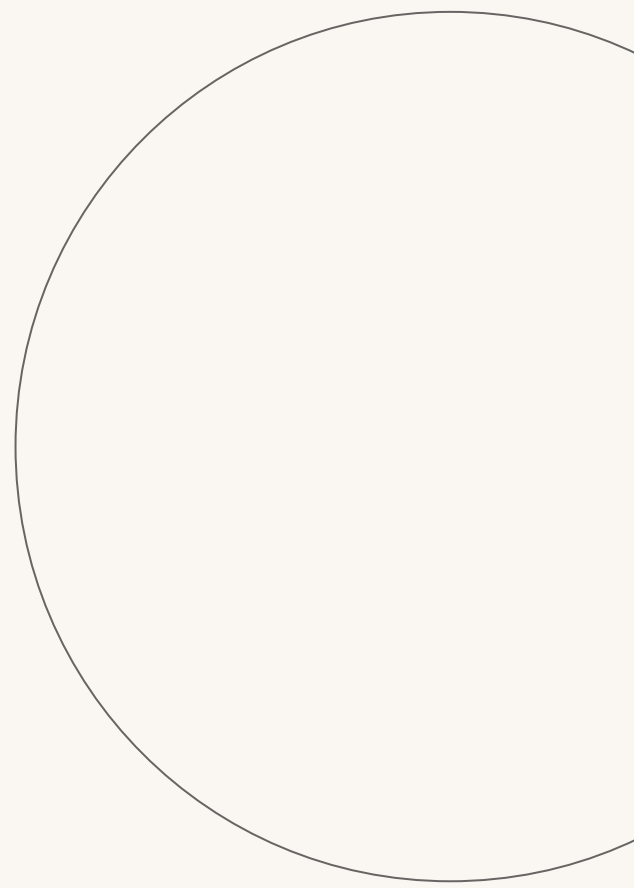
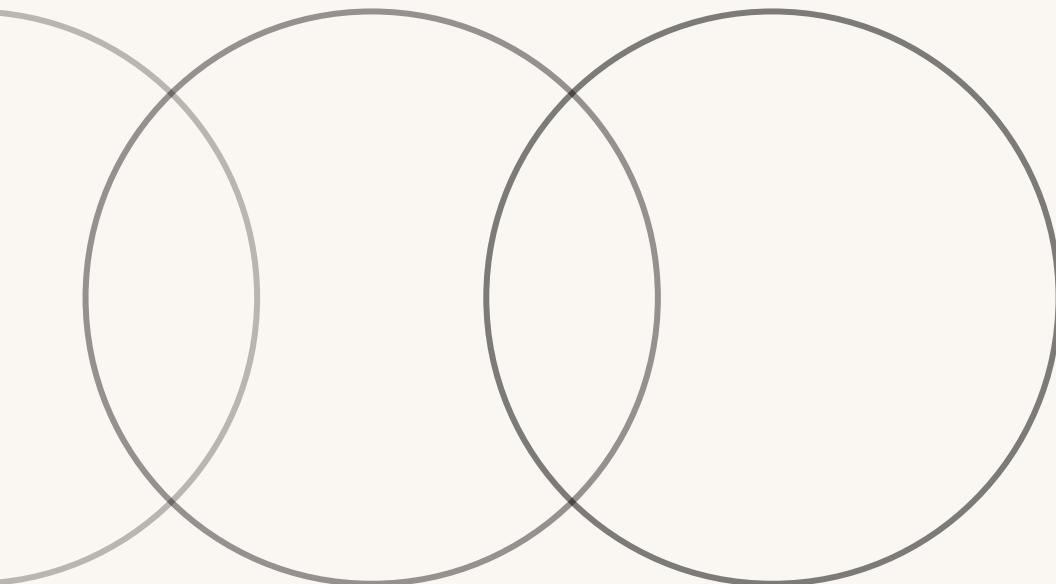
Group arm64

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Architecture Principles, Vision, and Requirements

Architectural Principles

- Security & Access Control
- Scalability & Performance
- Maintainability & Modularity
- Architecture Governance Compliance



Architecture Principles, Vision, & Requirements

Vision

Develop an integrated university platform that unifies SMAP, AUTHOR LMS, ePayment, eKursus, ePrestasi, and USEM under one system.

Driver: Need for centralized academic, financial, and student-related data.
Objective: Provide a single unified portal for all stakeholders.
Principle: Data accuracy and consistency across all systems.

Vision

To build a secure, scalable digital system that gives all users consistent web-based access anytime and anywhere

AS-IS

TO-BE

Constraints

Constraints

- The system must integrate all major UTHM systems under one platform.
- Must allow SSO authentication.

- Legacy systems (SMAP, ePrestasi, ePayment) must remain operational during transition.
- Budget limitations for large-scale migration.
- University policies regarding data privacy and government regulations.
- *No EA expertise within UTHM ICT Department.*

- The system must integrate all major UTHM systems under one platform.
- Must allow SSO authentication.
- **Support University KPI for digitalization.**

- Legacy systems (SMAP, ePrestasi, ePayment) must remain operational during transition.
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Requirements

Requirements

Business Architecture

AS-IS = TO-BE

Compenent	explanation		
Motivation	• <i>Improve student experience via seamless access to university services.</i> • <i>Reduce administrative workload and manual processing.</i> • <i>Integrate major systems (SMAP, LMS, ePayment, Library, ePrestasi) into one unified platform.</i>		
Organization	Organization Units	Locations	Actors / Roles
Function	• Core University Services • Service Quality		

Information System Architecture

Combination of data, application and services to support a business

AS-IS

Data

Data Entities

- Student
- Lecturer
- Courses
- Enrollments
- Performance records

Data Components

- Student Database
- Learning Material Database
- Master Database
- Data Recovery and back-up Database

Application

Information System Services (ISS)

- Identity & Authentication Service (SSO)
- Learning Service (AUTHOR Integration)
- Web-application based applications
- WAF and firewalls

Logical Application Components

- Student Portal Module
- Lecturer Portal Module
- Admin Management Module
- Payment Module
- Course Management Module
- Reporting & Analytics Module
- Workflow Automation Module

Physical Application Components

- Mobile & Web Frontend
- Legacy apps (TCIS) running on UTHM servers

TO-BE

Data

Data Entities

- Student
- Lecturer
- Courses
- Enrollment
- Payments
- Library Resources
- Events
- Performance Records

Logical Data Components

- Student Profile Database
- Course & Curriculum Database
- Financial Transaction Database
- Learning Content Data Store
- HR Performance Data Store

Physical Data Components

- On-prem databases for legacy systems (SMAP, ePrestasi, ePayment)
- Backup storage on another on-prem data centre

Application

Information System Services (ISS)

- Identity & Authentication Service (SSO)
- University Management Service (U-IMS)
- Academic Management Service (SMAP Integration)
- Financial Service (ePayment Integration)
- Learning Service (AUTHOR Integration)
- HR Appraisal Service (ePrestasi Integration)
- Event Management Service (USEM Integration)

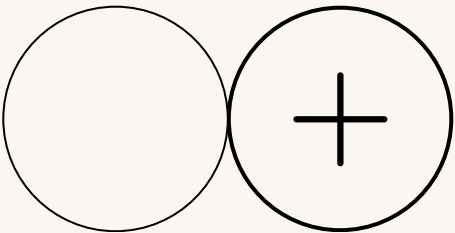
Logical Application Components

- Student Portal Module
- Lecturer Portal Module
- Admin Management Module
- Payment Module
- Course Management Module
- Reporting & Analytics Module
- Workflow Automation Module

Physical Application Components

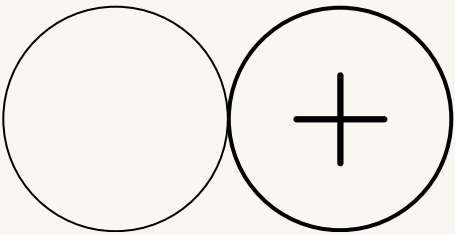
- API Integration Services
- Mobile & Web Frontend
- Legacy apps running on UTHM servers

Technology Architecture



Component	AS-IS	TO-BE
Infrastructure	- On-premise data centres (Batu Pahat and Pagoh) - SAN Storage Servers - Tier 3 Architecture - PDU and UPS power supply - Backup Generators - DCIM Software	- On-premise data centres (Batu Pahat and Pagoh) - SAN Storage Servers - Tier 3 Architecture - PDU and UPS power supply - Backup Generators - DCIM Software - Offsite recover
Platforms and Middleware	- Windows and Linux (Servers) - Web Application - iOS and Android (Mobile Devices) - MySQL (Relational DB) - Moodle (SaaS) - Payment Gateways - API	- Windows and Linux (Servers) - Web Application - iOS and Android (Mobile Devices) - MySQL (Relational DB) - MongoDB (NoSQL) - Moodle (SaaS) - Payment Gateways - VMware - API

Technology Architecture



Component	AS-IS	TO-BE
Network	• High speed LAN/WAN • Wi-Fi	• High speed LAN/WAN • Wi-Fi • VPN • Load Balancer
Security	• Single Sign-On (SSO) • HTTPS • Web Application Firewall (WAF) • Firewalls	• Single Sign-On (SSO) • HTTPS • Security Information and Event Management (SIEM) • Web Application Firewall (WAF) • Firewalls

Opportunities

Capabilities

AS-IS

- Student Information Management
- Academic & Teaching Support
- Research & Innovation Support
- Administrative & Support Services
- Data & Analytics Capability
- Digital Campus & IT Infrastructure

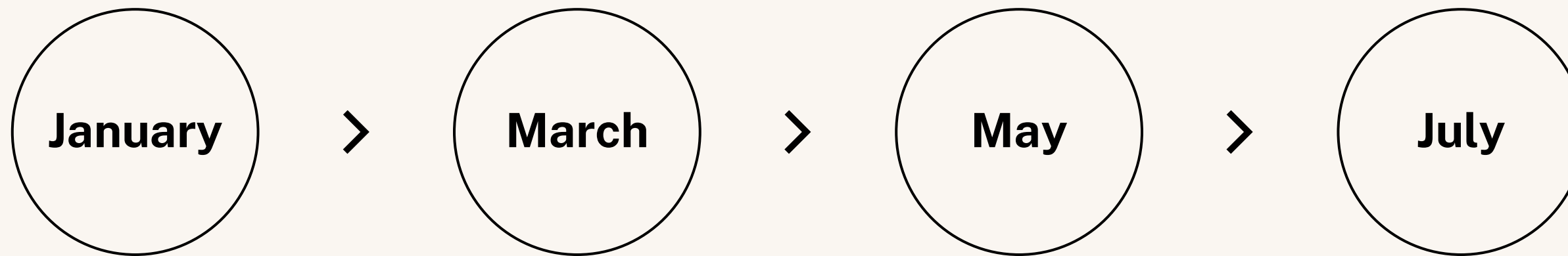
TO-BE

- Student Information Management
- Academic & Teaching Support
- Research & Innovation Support
- Administrative & Support Services
- Data & Analytics Capability
- Digital Campus & IT Infrastructure

Solutions

AS-IS	TO-BE
Existing Systems Analysis	Existing Systems Analysis
API Integration Development	API Integration Development
Migration of Selected Modules	Migration of Selected Modules
Development of U-IMS Portal	Development of U-IMS Portal
Testing & Training	Testing & Training

Migration Planning - Phase by Phase



- Migration is carried out incrementally, where each phase delivers part of the TO-BE architecture
- Migration takes around 3 years per module
- Migration Status: 70-80% done
- Multiple migration activities may overlap across phases
- If the legacy system cannot be migrate then a new system is to be-considered

Implementation Governance

EA AS-IS

- Financial System Request (FSR) approval process for new systems
- Jabatan Digital Negara (JDN) guidelines for website design and enterprise architecture
- Ministry of Higher Education (MOHE) for funding

EA TO-BE

STANDARDS

- Architectural Alignment: Every solution must strictly match the target Business, Data, Application, and Technology blueprints.
- Unified Integration: Systems are no longer allowed to work in isolation; they must use a common integration pattern (APIs).
- Compliance and Documentation: All projects must use university-approved templates for design and strictly follow data privacy laws

GUIDELINES

- Scalability & Reusability: Systems must be modular so they can grow. We prioritize reusable components to avoid wasting budget on building the same thing twice.
- Data Consistency: We establish shared data definitions across all departments to ensure a "Single Source of Truth".
- Unified User Experience (UX): A consistent look and feel across all U-IMS modules to make the system easy for students and staff to use.
- Security by Design: Implementing "Least Privilege" access, ensuring users only see the data necessary for their role.

SPECIFICATIONS

- Architectural Models: High-level diagrams and logical/physical models are required before any development begins.
- System Mapping: Detailed documentation of data flows, system interfaces, and integration points to prevent system errors.
- Operational Readiness: Clear plans for hosting (Cloud/On-premise), system monitoring, and disaster recovery to ensure 99% uptime.

The background features several thin, light gray circles of varying sizes. One circle is in the top left, a horizontal row of four overlapping circles is in the top right, a single circle is on the left side, and a large circle is in the bottom right. A small arc of a circle is visible at the bottom center.

Thank You